Data Structures and Algorithms for AI

NAME :
SURNAME :

FINAL
Duration : 2 hours

1. (10 points) What is Variance? (You may explain in plain english, or give the formula of variance, or explain it using an example)
2. (10 points) Given det(A.B)=det(A) . det(B), demonstrate that $det(A^{-1})=1/det(A)$ (No python code, only you will write maximum 3-4 lines of mathematical operations)
3. (10 points) Find the time complexity formula of the following algorithm as a function of n (data length).

```
def analyze_me(data):
    n = len(data)
    total = 0

    # Section 1: Linear Scan
    for i in range(n):
        total += data[i]

    # Section 2: Nested Loops
    for i in range(n):
        for j in range(0, n, 2):  # Note the step of 2
            total += (i * j)
```

4. (10 points) Compare the following complexity functions, and put them in order :
   I. O(n)
   II. O(n.log(n))
   III. $O(n^2)$
   IV. O(log(n))
5. (10 points) Write python code of the following object oriented design :
   I. A vehicle object that has
      1. An engine, four wheels, a chassis, a body, two front lights, two rear lights as attributes.
      2. accelerate(speed), decelerate(speed), set_direction(alfa,beta) as methods.
   II. A car is a vehicle and overrides the accelerate and decelarate methods.
   III. A truck is a vehicle and overrides the set_direction(alfa,beta) methods.
6. (10 points) Implement a binary tree class that has insert_node, remove_node as methods and “nodeValue” as the node value.
7. (10 points) Multiply the following matrices A and B,.(You have to transpose one of them).

$$A=\begin{bmatrix}1 & 0\\ 1 & 1\\ 0 & 1\end{bmatrix} \qquad B=\begin{bmatrix}2 & 1\\ 0 & 1\\ 1 & 0\end{bmatrix}$$

8. (20 points) Graph Definitions :
   I. Draw an undirected graph with 5 vertices each connected to at least 3 other vertices and each having a distinct integer number as value. Enumerate each vertex and write the corresponding number on the vertex.
   II. Write the degree of each vertex in the graph.
   III. Write the adjacency matrix of the graph.
   IV. What is a weighted graph, what should we do on the first graph to make it a “weighted graph”?

V. What would we do to make this graph acyclic (which edges would be removed keeping the graph connected) ?
VI. Give the definition of Breadth First Search. Suppose we search a number in the graph using this algorithm, what would be the first 3 vertex names/enumerations by starting the first vertex in the graph.

9. (20 points) Implement a graph class in python that has “insert_vertex”, “delete_vertex” and “breath_first_search” methods.